Chapter 8S. Learning Error and Neural Network Approximation

Supplementary handout: risk decomposition, curse of dimensionality, surrogate losses, universal approximation and Barron rates

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Remark 8S.1: about this chapter

This chapter rewrites a supplementary handout that organizes two post-midterm lectures. It asks three separate questions: why learning incurs error (Sections 1–3), whether a network can approximate a target (Sections 4–5), and how the approximation error scales with the width (Sections 6–7). Sections 4–7 overlap with Chapter 8; here they are kept short and cross-referenced, while the new material on risk, sample size and surrogate losses is treated in full.

1 Risk and the Sources of Learning Error

Motivation. A trained network is judged on new data, but it is fitted on a finite sample with an imperfect optimizer inside a restricted model class. Each of these three restrictions costs something, and the excess risk splits accordingly.

1.1 Population and empirical risk

Definition 8S.2: population and empirical risk

Let $S = ((X_i, Y_i))_{i=1}^n \sim P^{\otimes n}$ be a training sample and ℓ a loss. For a predictor f,

$$R(f) = \mathbb{E}_{(X,Y)\sim P}\, \ell(f(X), Y), \qquad \widehat{R}_S(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(X_i), Y_i). \tag{1}$$

R is the **population risk**, $\widehat{R}_S$ the **empirical risk**. The expectation in R is over a fresh observation, not over the training sample.

Let $\mathcal{F}_\delta \subseteq \mathcal{F}$ be a complexity-constrained class and $\widehat{f}_S \in \mathcal{F}_\delta$ the output of a learning algorithm. Assume all risks and infima below are finite. Then, by adding and subtracting $\inf_{\mathcal{F}_\delta} R$,

$$R(\widehat{f}_S) - \inf_{f\in\mathcal{F}} R(f) = \underbrace{\inf_{f\in\mathcal{F}_\delta} R(f) - \inf_{f\in\mathcal{F}} R(f)}_{\varepsilon_{\text{app}}} + \underbrace{R(\widehat{f}_S) - \inf_{f\in\mathcal{F}_\delta} R(f)}_{\varepsilon_{\text{est}}}. \tag{2}$$

The **approximation error** ε_{app} is the price of restricting the class; the **estimation error** ε_{est} measures how far the learned predictor is from the best predictor in that class. Both are nonnegative: $\mathcal{F}_\delta \subseteq \mathcal{F}$ gives $\inf_{\mathcal{F}_\delta} R \geq \inf_{\mathcal{F}} R$, and $\widehat{f}_S \in \mathcal{F}_\delta$ gives $R(\widehat{f}_S) \geq \inf_{\mathcal{F}_\delta} R$.

Definition 8S.3: generalization gap and uniform deviation

$$\mathrm{gen}_S(f) = R(f) - \widehat{R}_S(f), \qquad \Delta_S(\mathcal{F}_\delta) = \sup_{f\in\mathcal{F}_\delta} |R(f) - \widehat{R}_S(f)|. \tag{3}$$

The gap is signed and belongs to one predictor. The uniform deviation is nonnegative and controls every member of the class at once; in particular $|\mathrm{gen}_S(\widehat{f}_S)| \leq \Delta_S(\mathcal{F}_\delta)$ because $\widehat{f}_S \in \mathcal{F}_\delta$.

Remark 8S.4: notation

The lecture writes ε_{gen} for a supremum term, and other sources use the same symbol for an individual gap. These notes keep the two apart (gen_S versus Δ_S). Throughout, n is the sample size, d the input dimension and N the hidden-layer width.

1.2 Optimization error and a uniform bound

Definition 8S.5: optimization error

$$\varepsilon_{\text{opt}} = \widehat{R}_S(\widehat{f}_S) - \inf_{f\in\mathcal{F}_\delta} \widehat{R}_S(f) \;\geq 0. \tag{4}$$

It is zero exactly when the algorithm returns an empirical risk minimizer. Adding and subtracting $\widehat{R}_S(\widehat{f}_S)$ and $\inf_{\mathcal{F}_\delta} \widehat{R}_S$ gives the exact identity

$$\varepsilon_{\text{est}} = \text{gen}_S(\widehat{f}_S) + \varepsilon_{\text{opt}} + \Big(\inf_{f\in\mathcal{F}_\delta} \widehat{R}_S(f) - \inf_{f\in\mathcal{F}_\delta} R(f) \Big). \tag{5}$$

The first and third terms can be negative, and the third cannot be discarded in general.

Theorem 8S.6: excess-risk bound

For $\widehat{f}_S \in \mathcal{F}_\delta$,

$$R(\widehat{f}_S) - \inf_{f\in\mathcal{F}} R(f) \leq \varepsilon_{\text{app}} + 2\Delta_S(\mathcal{F}_\delta) + \varepsilon_{\text{opt}}.$$

Proof. Write $\Delta = \Delta_S(\mathcal{F}_\delta)$.

Step 1 (two-sided comparison). By definition of Δ, for every $f \in \mathcal{F}_\delta$,

$$R(f) - \Delta \leq \widehat{R}_S(f) \leq R(f) + \Delta.$$

Step 2 (the infimum term). Taking the infimum over $f \in \mathcal{F}_\delta$ in each inequality of Step 1 gives $\inf R - \Delta \leq \inf \widehat{R}_S \leq \inf R + \Delta$, so the last term of (5) lies in $[-\Delta, \Delta]$.

Step 3 (the gap term). $\text{gen}_S(\widehat{f}_S) \leq |\text{gen}_S(\widehat{f}_S)| \leq \Delta$.

Step 4. Steps 2 and 3 in (5) give $\varepsilon_{\text{est}} \leq 2\Delta + \varepsilon_{\text{opt}}$; insert this into (2). ■

The three terms on the right are exactly the three sources of error of Chapter 8, Section 1: approximation, generalization (through Δ) and optimization.

1.3 A fixed predictor versus a learned predictor

Theorem 8S.7: fixed predictor

Fix f before seeing the sample and let $Z_i = \ell(f(X_i), Y_i)$ have variance $0 < \sigma_f^2 < \infty$. Then

$$\mathbb{E}_S[\widehat{R}_S(f) - R(f)] = 0, \qquad \text{Var}_S\left(\widehat{R}_S(f) - R(f)\right) = \frac{\sigma_f^2}{n},$$

and for every $c > 0$

$$P_S\Big(|\widehat{R}_S(f) - R(f)| \geq \frac{c\,\sigma_f}{\sqrt{n}}\Big) \leq \frac{1}{c^2}.$$

Hence $\widehat{R}_S(f) - R(f) = O_P(n^{-1/2})$. If $\sigma_f = 0$, the difference is 0 almost surely.

Proof. Step 1 (iid losses). Since f does not depend on S, the Z_i are iid with mean $\mathbb{E}Z_i = R(f)$ and variance σ_f^2.

Step 2 (mean and variance). $\widehat{R}_S(f) = \frac{1}{n}\sum_i Z_i$ has mean $R(f)$ and, by independence, variance $\frac{1}{n^2} \cdot n\sigma_f^2 = \sigma_f^2/n$. Subtracting the constant $R(f)$ changes neither the variance nor the zero mean of the difference.

Step 3 (Chebyshev). For a random variable D with mean 0 and variance v, $P(|D| \geq t) \leq v/t^2$. With $v = \sigma_f^2/n$ and $t = c\sigma_f/\sqrt{n}$ the bound is $1/c^2$.

Step 4 ($\sigma_f = 0$). Then $Z_i = R(f)$ almost surely for every i, so $\widehat{R}_S(f) = R(f)$ almost surely. ■

This calculation does not transfer to $\widehat{f}_S$: the losses $\ell(\widehat{f}_S(X_i), Y_i)$ are computed with a predictor chosen from the same sample, so they are neither independent nor centred at $R(\widehat{f}_S)$.

Example 8S.8: a memorizing predictor [Supplement]

Let X be uniform on $[0,1]$ and $Y \in \{-1,1\}$ a fair coin independent of X; use the 0–1 loss. Every predictor has $R(f) = \frac{1}{2}$. Let $\widehat{f}_S(x) = Y_i$ if $x = X_i$ and $\widehat{f}_S(x) = 1$ otherwise. Then $\widehat{R}_S(\widehat{f}_S) = 0$, while a fresh X almost surely differs from every X_i, so $R(\widehat{f}_S) = P(Y \neq 1) = \frac{1}{2}$. The gap $\mathrm{gen}_S(\widehat{f}_S) = \frac{1}{2}$ does not shrink with n, although every *fixed* f has gap $O_P(n^{-1/2})$ by Theorem 8S.7. Uniform control $\Delta_S(\mathcal{F}_\delta)$ is what rules this out, and it fails here because the class allowed to $\widehat{f}_S$ is too rich.

Expectation and high-probability statements. The lecture contrasts guarantees of the form

$$\mathbb{E}_S|\mathrm{gen}_S(\widehat{f}_S)| \leq \varepsilon \qquad \text{and} \qquad P_S\{|\mathrm{gen}_S(\widehat{f}_S)| \leq \varepsilon\} \geq 1 - \eta.$$

These are *forms* of guarantees, not consequences that hold without assumptions. The second is the PAC (probably approximately correct) viewpoint; a full PAC definition also fixes the risk target and how n must grow. A small *signed* expectation $\mathbb{E}_S\,\mathrm{gen}_S(\widehat{f}_S)$ alone implies neither bound, since positive and negative gaps can cancel.

Theorem 8S.9: complexity trade-off for nested classes

If $\mathcal{F}_\delta \subseteq \mathcal{F}_{\delta'} \subseteq \mathcal{F}$, then $\varepsilon_{\mathrm{app}}(\mathcal{F}_{\delta'}) \leq \varepsilon_{\mathrm{app}}(\mathcal{F}_\delta)$ and $\Delta_S(\mathcal{F}_{\delta'}) \geq \Delta_S(\mathcal{F}_\delta)$.

Proof. An infimum over a larger set is no larger, so $\inf_{\mathcal{F}_{\delta'}} R \leq \inf_{\mathcal{F}_\delta} R$; subtract $\inf_{\mathcal{F}} R$. A supremum over a larger set is no smaller, which gives the second inequality. ■

Enlarging the class therefore trades approximation error against uniform deviation. This is a statement about the two bounds, not a claim that every observed test-error curve has a particular shape.

2 The Curse of Dimensionality

Motivation. Theorem 8S.6 does not say how much data is needed. The answer depends heavily on how structured the target is: a linear target needs about d samples, a merely Lipschitz target needs a number exponential in d.

Example 8S.10: noiseless linear recovery

Let $y_i = \mathbf{x}_i^\top \boldsymbol{\theta}^*$ with design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ (rows $\mathbf{x}_i^\top$). Then $\boldsymbol{\theta}^*$ is uniquely determined by the data if and only if $\operatorname{rank} \mathbf{X} = d$. In particular $n \geq d$ is necessary, but not sufficient by itself.

Proof. Step 1 (sufficiency). If $\operatorname{rank} \mathbf{X} = d$, the columns are independent, so $\mathbf{X}\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\theta}'$ implies $\boldsymbol{\theta} = \boldsymbol{\theta}'$; the system $\mathbf{X}\boldsymbol{\theta} = \mathbf{y}$ has the single solution $\boldsymbol{\theta}^*$.

Step 2 (necessity). If $\operatorname{rank} \mathbf{X} < d$, there is $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$, and $\boldsymbol{\theta}^* + \mathbf{v}$ fits the data equally well.

Step 3. $\operatorname{rank}\mathbf{X} \le \min(n, d)$, so rank d needs $n \ge d$. Repeating one row n times shows that $n \ge d$ alone does not give rank d. ■

Theorem 8S.11: Gaussian designs have full rank [Supplement]

If the rows $\mathbf{x}_i$ are iid $\mathcal{N}(\mathbf{0}, \Sigma)$ with Σ positive definite and $n \ge d$, then $\operatorname{rank}\mathbf{X} = d$ almost surely.

Proof. Step 1. It suffices that the top $d \times d$ block $\mathbf{X}_d$ is invertible, i.e. $\det \mathbf{X}_d \neq 0$.

Step 2. $\det \mathbf{X}_d$ is a polynomial in the d^2 entries and is not identically zero (it equals 1 at the identity). The zero set of a nonzero polynomial on $\mathbb{R}^{d^2}$ has Lebesgue measure zero.

Step 3. Because Σ is positive definite, the entries of $\mathbf{X}_d$ have a joint density on $\mathbb{R}^{d^2}$, so they fall in a Lebesgue-null set with probability 0. ■

2.1 Lipschitz targets and nearest neighbours

Now let $y_i = f^*(\mathbf{x}_i)$ with $\operatorname{Lip}(f^*) \le \beta$, where $\operatorname{Lip}(f) = \sup_{\mathbf{x}\neq\mathbf{x}'} |f(\mathbf{x}) - f(\mathbf{x}')|/\|\mathbf{x} - \mathbf{x}'\|$. Assume a minimum-Lipschitz interpolant exists:

$$\widehat{f} \in \arg\min \{\operatorname{Lip}(f) : \ f(\mathbf{x}_i) = y_i \text{ for all } i\}. \tag{6}$$

Theorem 8S.12: error controlled by the nearest observation

If $\mathbf{x}_k$ is a training point nearest to $\mathbf{x}$, then $|\widehat{f}(\mathbf{x}) - f^*(\mathbf{x})| \le 2\beta\|\mathbf{x} - \mathbf{x}_k\|$.

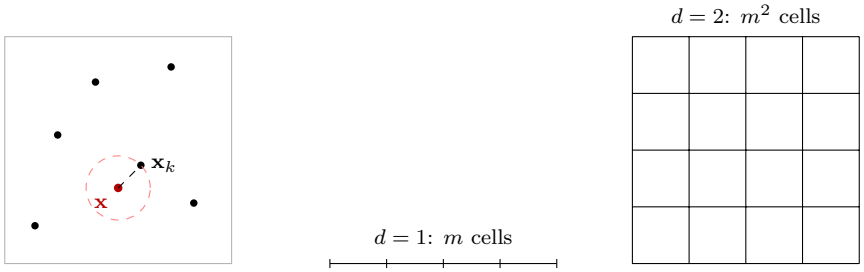
Proof. Step 1. f^* is feasible in (6), so $\operatorname{Lip}(\widehat{f}) \le \operatorname{Lip}(f^*) \le \beta$.

Step 2. Since $\widehat{f}(\mathbf{x}_k) = y_k = f^*(\mathbf{x}_k)$,

$$|\widehat{f}(\mathbf{x}) - f^*(\mathbf{x})| \le |\widehat{f}(\mathbf{x}) - \widehat{f}(\mathbf{x}_k)| + |f^*(\mathbf{x}_k) - f^*(\mathbf{x})| \le \beta\|\mathbf{x} - \mathbf{x}_k\| + \beta\|\mathbf{x} - \mathbf{x}_k\|.$$

■

The key quantity is therefore the distance to the nearest observation. Dividing each coordinate of $[0, 1]^d$ into m intervals creates m^d cells; to have one observation in each cell, i.e. resolution $1/m$ everywhere, needs $n \approx m^d$ samples, so $\log n \approx d \log m$ grows linearly in d at fixed resolution.



(a) local coverage (b) fixed resolution $1/m$

Local interpolation needs nearby observations; keeping the same resolution costs m^d cells.

Theorem 8S.13: nearest-neighbour distance in the unit cube [Supplement]

Let $X, X_1, \ldots, X_n$ be iid uniform on $[0, 1]^d$ and $D = \min_i \|X - X_i\|$. Then

$$\mathbb{E}\, D^2 \le C_d\, n^{-2/d}, \qquad C_d = d\,\Gamma(1 + \tfrac{2}{d}).$$

Consequently $\mathbb{E}|\widehat{f}(X) - f^*(X)|^2 \le 4\beta^2 C_d n^{-2/d}$, and this bound is at most ε once $n \ge (4\beta^2 C_d/\varepsilon)^{d/2}$.

Proof. Step 1 (small-ball probability). Fix $\mathbf{x} \in [0,1]^d$ and $0 < r \le \sqrt{d}$. The box $\{\mathbf{u} \in [0,1]^d : |u_j - x_j| \le r/\sqrt{d}\ \forall j\}$ lies in the ball of radius r around $\mathbf{x}$ (since $\|\mathbf{u} - \mathbf{x}\| \le \sqrt{d}\max_j |u_j - x_j|$), and each of its sides has length at least $r/\sqrt{d}$ because $r/\sqrt{d} \le 1$. Hence

$$p(\mathbf{x}, r) := P(\|X_1 - \mathbf{x}\| \le r) \ge d^{-d/2} r^d.$$

Step 2 (tail of D). Given $X = \mathbf{x}$, the events $\|X_i - \mathbf{x}\| > r$ are independent, so with $1 - p \le e^{-p}$

$$P(D > r \mid X = \mathbf{x}) = (1 - p(\mathbf{x}, r))^n \le \exp\big(-n\, d^{-d/2} r^d\big) \qquad (0 < r \le \sqrt{d}).$$

For $r > \sqrt{d}$ the left side is 0, because the diameter of the cube is $\sqrt{d}$; so the bound holds for all $r > 0$.

Step 3 (integrate the tail). With $\mathbb{E}[D^2 \mid X] = \int_0^\infty P(D^2 > s \mid X)\, ds$,

$$\mathbb{E}[D^2 \mid X] \le \int_0^\infty \exp\big(-n\, d^{-d/2} s^{d/2}\big)\, ds.$$

Substitute $u = n\, d^{-d/2} s^{d/2}$, i.e. $s = d\,(u/n)^{2/d}$ and $ds = d\, n^{-2/d} \frac{2}{d}\, u^{2/d-1} du$:

$$\int_0^\infty e^{-u}\, d\, n^{-2/d} \tfrac{2}{d}\, u^{2/d-1}\, du = d\, n^{-2/d}\, \tfrac{2}{d}\, \Gamma\big(\tfrac{2}{d}\big) = d\, \Gamma\big(1 + \tfrac{2}{d}\big)\, n^{-2/d}.$$

Take the expectation over X.

Step 4 (prediction error). Square Theorem 8S.12 with $\mathbf{x} = X$, take expectations, and solve $4\beta^2 C_d n^{-2/d} \le \varepsilon$ for n. ■

To certify squared error ε through this bound needs n of order $\varepsilon^{-d/2}$; for squared error ε^2 the order is ε^{-d}.

Remark 8S.14: upper versus lower bounds

The rate needs distributional assumptions (here uniform inputs). An upper bound supplies a *sufficient* sample size; by itself it does not prove that every algorithm needs that many samples.

3 Surrogate Loss Is Not Classification Error

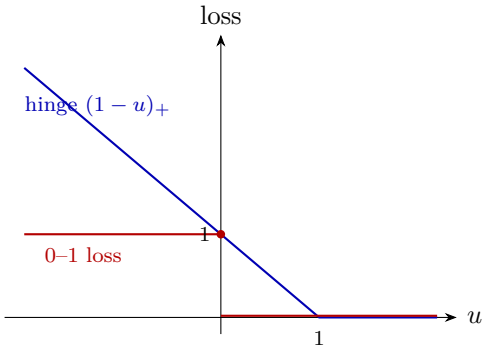
Motivation. All terms in Section 1 refer to one fixed loss. When that loss is a convenient surrogate such as the hinge loss, small risk for the surrogate does not single out the classifier with the fewest mistakes.

Let $Y \in \{-1, 1\}$ and let $g : \mathcal{X} \to \mathbb{R}$ be a score with prediction $\operatorname{sign} g$. The **signed margin** is $u = Yg(X)$, and

$$\ell_{\text{hinge}}(u) = (1 - u)_+ = \max\{0, 1 - u\}, \qquad \mathbf{1}\{u \le 0\} \le (1 - u)_+. \tag{7}$$

Proof of the inequality. If $u \le 0$, then $1 - u \ge 1 = \mathbf{1}\{u \le 0\}$. If $u > 0$, the left side is $0 \le (1 - u)_+$. ■

Counting zero margins as errors makes the left side the classification loss; with other tie-breaking rules the misclassification indicator is still at most $\mathbf{1}\{u \le 0\}$, hence at most the hinge loss.



Hinge loss also penalizes correct predictions with margin below 1*; the* 0–1 *loss only records mistakes.*

Example 8S.15: different rankings

Let $P(X = x_1) = 0.99$, $P(X = x_2) = 0.01$, and $Y = +1$ at both points. Compare two scores:

	$g(x_1)$	$g(x_2)$	$R_{0\text{–}1}$	R_{hinge}
g_A	0.1	0.1	0	0.9
g_B	1	-0.1	0.01	0.011

Here g_A makes no mistakes, but hinge risk strongly prefers g_B.

Computation. g_A: both margins are $0.1 > 0$, so no errors, and each hinge loss is 0.9; $R_{\text{hinge}}(g_A) = 0.9$. g_B: at x_1 the margin is 1, hinge 0, correct; at x_2 the margin is -0.1, hinge 1.1, wrong. So $R_{0\text{–}1}(g_B) = 0.01$ and $R_{\text{hinge}}(g_B) = 0.99 \cdot 0 + 0.01 \cdot 1.1 = 0.011$. ■

The first score is always correct with a small margin; the second sacrifices a rare point to gain a comfortable margin on most inputs.

Remark 8S.16: no contradiction

Both scores satisfy $R_{0\text{–}1}(g) \le R_{\text{hinge}}(g)$ ($0 \le 0.9$ and $0.01 \le 0.011$). A smaller upper bound need not sit above a smaller quantity. If the loss in Section 1 is the hinge loss, controlling approximation, sampling and optimization controls hinge risk; it does not by itself identify the class member with the smallest classification error. This two-candidate example does not refute general classification-calibration results, which concern minimizers over all measurable scores.

4 Universal Approximation

Motivation. We turn from statistical error to representation: can one hidden layer approximate any continuous target at all? This controls ε_{app} in (2).

Let $K = [0,1]^d$ and $C(K) = \{f : K \to \mathbb{R} \text{ continuous}\}$ with $\|f\|_\infty = \sup_{\mathbf{x}\in K} |f(\mathbf{x})|$, a Banach space. A subset U is **dense** if every $f \in C(K)$ is approximated arbitrarily closely in this norm by elements of U.

Definition 8S.17: sigmoidal activation

A continuous $\sigma : \mathbb{R} \to [0,1]$ is **sigmoidal** if $\lim_{t\to-\infty} \sigma(t) = 0$ and $\lim_{t\to+\infty} \sigma(t) = 1$.

The logistic function is one example, not the only one. The handwritten definition also asks for C^1 regularity; continuity suffices below.

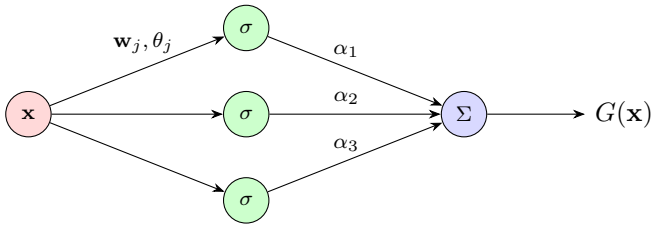
Theorem 8S.18: Cybenko 1989

For a fixed continuous sigmoidal σ, the class

$$\mathcal{G}_\sigma = \Big\{ \mathbf{x} \mapsto \sum_{j=1}^{N} \alpha_j \sigma(\mathbf{w}_j^\top \mathbf{x} + \theta_j) : \ N \in \mathbb{N},\ \alpha_j, \theta_j \in \mathbb{R},\ \mathbf{w}_j \in \mathbb{R}^d \Big\}$$

is dense in $(C(K), \|\cdot\|_\infty)$: for every $f \in C(K)$ and $\varepsilon > 0$ there is $G \in \mathcal{G}_\sigma$ with $\sup_{\mathbf{x}\in K} |f(\mathbf{x}) - G(\mathbf{x})| < \varepsilon$.

The proof is in Section 5 (and in Chapter 8, Theorem 8.4).



One hidden layer with three units. "Two-layer" counts the hidden and output layers; the theorem lets the width vary.

Reading the quantifiers. The width N and all weights may depend on f and ε. The theorem is not a fixed-width statement, gives no useful bound on N, and provides no algorithm that finds the weights.

Theorem 8S.19: link to approximation error

Let $f^* \in C(K)$, $Y = f^*(X)$ with X supported in K, and squared loss. If $\sup_K |f^* - G| < \varepsilon$, then $R(G) = \mathbb{E}(G(X) - f^*(X))^2 < \varepsilon^2$. Hence $\varepsilon_{\text{app}}(\mathcal{G}_\sigma) = 0$ for this problem.

Proof. $(G(X) - f^*(X))^2 < \varepsilon^2$ holds pointwise on K; take expectations. Since $R \geq 0$ and $\inf_{\mathcal{F}} R = R(f^*) = 0$, letting $\varepsilon \to 0$ gives $\inf_{\mathcal{G}_\sigma} R = 0$. ■

The implication uses the chosen loss; it is not automatic for arbitrary losses.

5 The Functional-Analytic Proof

Motivation. Cybenko's proof never constructs weights. It assumes density fails, turns the failure into a nonzero measure that "sees" no neuron, and shows that no such measure exists.

Let $M(K)$ be the finite regular signed Borel measures on K and $|\mu|$ the total variation measure. Integration against μ is a continuous linear functional because

$$\Big| \int_K h \, d\mu \Big| \leq \|h\|_\infty \, |\mu|(K). \tag{8}$$

The measure need not be positive or a probability measure.

Definition 8S.20: discriminatory activation

σ is **discriminatory** on K if for every $\mu \in M(K)$

$$\int_K \sigma(\mathbf{w}^\top \mathbf{x} + \theta) \, d\mu(\mathbf{x}) = 0 \quad \text{for all } \mathbf{w}, \theta \qquad \Longrightarrow \qquad \mu = 0.$$

Theorem 8S.21: two separation tools

(i) If a linear subspace U is not dense in a normed space X, there is a bounded linear functional $L \neq 0$ on X with $L|_U = 0$.

(ii) (Riesz) Every bounded linear functional L on $C(K)$ has a unique representation $L(h) = \int_K h\, d\mu$ with $\mu \in M(K)$, and $\|L\| = |\mu|(K)$.

Proof. (i) [Supplement] (the lecture omits it; the argument is Chapter 8, Theorem 8.5). The closure $\overline{U}$ is a proper closed subspace, so pick $x_0 \notin \overline{U}$ with $\rho = \inf_{u \in \overline{U}} \|x_0 - u\| > 0$. On $\overline{U} \oplus \mathbb{R}x_0$ define $\ell(u + tx_0) = t$; it vanishes on U, $\ell(x_0) = 1$, and $|t|\rho \leq \|u + tx_0\|$ shows $\|\ell\| \leq 1/\rho$. Hahn–Banach extends ℓ to X with the same norm.

(ii) This is the Riesz representation theorem, quoted without proof as in the lecture. ■

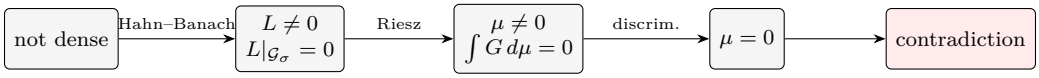
Theorem 8S.22: discriminatory activations give density

If σ is continuous and discriminatory, then $\mathcal{G}_\sigma$ is dense in $C(K)$.

Proof. Step 1 (subspace). Continuity of σ gives $\mathcal{G}_\sigma \subseteq C(K)$. Finite sums can be concatenated and scalars absorbed into the α_j, so $\mathcal{G}_\sigma$ is a linear subspace; this uses the freedom to vary N.

Step 2 (a nonzero measure). Suppose $\mathcal{G}_\sigma$ is not dense. Theorem 8S.21(i) gives $L \neq 0$ with $L|_{\mathcal{G}_\sigma} = 0$, and (ii) gives $\mu \in M(K)$ with $L(h) = \int h\, d\mu$; $\mu \neq 0$ because $L \neq 0$.

Step 3 (contradiction). A single hidden unit lies in $\mathcal{G}_\sigma$, so $\int_K \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$. Discrimination forces $\mu = 0$. ■



The proof is a separation argument, not a construction of weights.

Theorem 8S.23: continuous sigmoidals are discriminatory [Supplement]

Every continuous sigmoidal σ is discriminatory on K.

The lecture states this lemma without proof; with it, Theorem 8S.22 proves Cybenko's theorem. The following argument fills the gap (compare Chapter 8, Theorem 8.9).

Proof. Assume $\int_K \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$.

Step 1 (sharpening). Fix $\mathbf{w}, \theta, \varphi$ and let $\Pi = \{\mathbf{x} \in K : \mathbf{w}^\top \mathbf{x} + \theta > 0\}$ (open half-space) and $H = \{\mathbf{x} \in K : \mathbf{w}^\top \mathbf{x} + \theta = 0\}$ (hyperplane). As $\lambda \to \infty$,

$$\sigma(\lambda(\mathbf{w}^\top \mathbf{x} + \theta) + \varphi) \longrightarrow \begin{cases} 1, & \mathbf{x} \in \Pi, \\ \sigma(\varphi), & \mathbf{x} \in H, \\ 0, & \text{otherwise.} \end{cases}$$

Each function on the left is a single neuron (weights $\lambda\mathbf{w}$, bias $\lambda\theta + \varphi$), so its integral is 0.

Step 2 (limit). The functions are bounded by 1 and $|\mu|(K) < \infty$, so dominated convergence (applied to μ^+ and μ^-) gives

$$0 = \mu(\Pi) + \sigma(\varphi)\, \mu(H) \qquad \text{for every } \varphi.$$

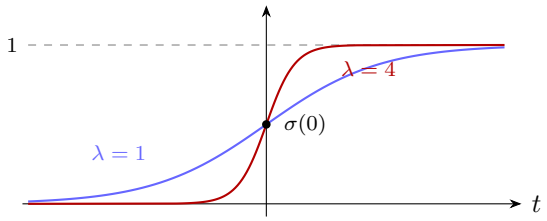
Step 3 (separate the two terms). Letting $\varphi \to -\infty$ gives $\mu(\Pi) = 0$; then $\varphi \to +\infty$ gives $\mu(H) = 0$.

Step 4 (pushforward). Fix $\mathbf{w}$ and let $\nu_{\mathbf{w}}(B) = \mu(\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} \in B\})$, a finite signed Borel measure on $\mathbb{R}$. By Step 3 every open half-line (a, ∞) and every point have $\nu_{\mathbf{w}}$-measure 0, hence so does every interval $(a, b] = (a, \infty) \setminus (b, \infty)$. Half-open intervals together with $\mathbb{R}$ form a π-system generating the Borel sets, and the finite measures $\nu_{\mathbf{w}}^+$, $\nu_{\mathbf{w}}^-$ agree on it; by the π–λ theorem they agree everywhere, so $\nu_{\mathbf{w}} = 0$.

Step 5 (Fourier). For all $t \in \mathbb{R}$ and $\mathbf{w}$,

$$\int_K e^{it\mathbf{w}^\top \mathbf{x}} \, d\mu(\mathbf{x}) = \int_{\mathbb{R}} e^{its} \, d\nu_{\mathbf{w}}(s) = 0.$$

The Fourier transform of the finite signed measure μ vanishes identically, and uniqueness of Fourier transforms gives $\mu = 0$. ■



For the logistic sigmoid, $\sigma(\lambda t)$ sharpens toward a step away from $t = 0$; the value at $t = 0$ stays $\sigma(0)$.

Remark 8S.24: two technical points

(1) Do not choose a finite φ with $\sigma(\varphi) = 0$; for the logistic sigmoid no such point exists. Use the limit $\varphi \to -\infty$. (2) Do not assume the hyperplane H has μ-measure zero; for a signed measure it may carry mass (e.g. a point mass). The proof keeps the term $\sigma(\varphi)\mu(H)$ and shows it vanishes. The handout names the half-space H and the hyperplane Π; these notes follow Chapter 8 (Π half-space, H hyperplane). On $[0, 1]^d$ the Baire and Borel σ-algebras coincide, so Borel terminology is used throughout.

6 Fourier Regularity and Barron Classes

Motivation. Universal approximation is qualitative. A rate in the width N needs extra structure on the target, measured through its Fourier transform.

Write $\widehat{f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \mathbf{x}^\top \boldsymbol{\xi}} \, d\mathbf{x}$. In this section and the next, the hat denotes a Fourier transform, not a learned predictor.

Definition 8S.25: Sobolev space

For $s \geq 0$,

$$H^s(\mathbb{R}^d) = \Big\{ f \in L^2(\mathbb{R}^d) : \int_{\mathbb{R}^d} (1 + \|\boldsymbol{\xi}\|^2)^s |\widehat{f}(\boldsymbol{\xi})|^2 \, d\boldsymbol{\xi} < \infty \Big\},$$

with the Fourier transform taken in the L^2 sense. In particular $H^0 = L^2$.

Larger s penalizes high frequencies more. Barron-type assumptions replace this weighted L^2 control by weighted L^1 control.

Definition 8S.26: Barron class (lecture version)

For $C_1 > 0$,

$$\Gamma_{C_1} = \Big\{ f \in L^1(\mathbb{R}^d) : \ \widehat{f} \in L^1(\mathbb{R}^d), \ \int_{\mathbb{R}^d} 2\pi\|\boldsymbol{\xi}\| \, |\widehat{f}(\boldsymbol{\xi})| \, d\boldsymbol{\xi} \le C_1 \Big\}.$$

Targets are real valued and taken as the continuous representatives given by Fourier inversion.

For a function defined only on a domain Ω, the **spectral Barron norm** uses extensions:

$$\|f\|_{\mathcal{B}^s(\Omega)} = \inf_{f_e|_\Omega = f} \int_{\mathbb{R}^d} (1 + \|\boldsymbol{\xi}\|)^s |\widehat{f_e}(\boldsymbol{\xi})| \, d\boldsymbol{\xi},$$

over admissible whole-space extensions f_e for which the right side is defined; the space consists of the f with finite norm. Extensions agree with f on Ω but may differ outside, and the norm picks the cheapest one.

Remark 8S.27: convention check

The weights $2\pi\|\boldsymbol{\xi}\|$ and $(1 + \|\boldsymbol{\xi}\|)^s$ differ, so Γ_{C_1} is not a ball of $\mathcal{B}^1$ with the same constant without further argument. Chapter 8 defines Γ_C with a strict inequality $< C$; the non-strict version here changes nothing in Theorem 8S.28 (see its proof). Admissible-extension conventions also vary between references.

7 A Quantitative Approximation Theorem

Motivation. Barron's theorem turns the qualitative density statement into an explicit rate in the width N, with an exponent independent of d.

Let $B_1^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le 1\}$ and

$$\|h\|^2_{L^2(\mathrm{Unif}(B_1^d))} = \frac{1}{\mathrm{Leb}_d(B_1^d)} \int_{B_1^d} |h(\mathbf{x})|^2 \, d\mathbf{x}.$$

Theorem 8S.28: Barron approximation, lecture statement

Let $f \in \Gamma_{C_1}$ and let σ be continuous and sigmoidal. For every $c > 4C_1^2$ and $N \in \mathbb{N}$ there is a network

$$\Phi_N(\mathbf{x}) = a_0 + \sum_{j=1}^{N} a_j \, \sigma(\mathbf{w}_j^\top \mathbf{x} + b_j)$$

with N hidden units and an output constant a_0 such that $\|f - \Phi_N\|^2_{L^2(\mathrm{Unif}(B_1^d))} < c/N$.

Proof. Step 1 (reduce to Chapter 8). Since $c > 4C_1^2$, choose C with $C_1 < C < \sqrt{c}/2$. Then $\int 2\pi\|\boldsymbol{\xi}\||\widehat{f}| \, d\boldsymbol{\xi} \le C_1 < C$, so f belongs to the class Γ_C of Chapter 8, and $c > 4C^2$.

Step 2 (apply the full proof). Barron's theorem as proved in Chapter 8 (Theorem 8.12, via Maurey's Lemma 8.15 and Lemma 8.16) gives, for this C and c, a network $f_N(\mathbf{x}) = f(\mathbf{0}) + \sum_{i=1}^N \frac{\gamma_i}{N} \sigma(\lambda \mathbf{a}_i^\top \mathbf{x} + \lambda b_i)$ with mean squared error $< c/N$. It has the stated form with $a_0 = f(\mathbf{0})$, $a_j = \gamma_j/N$, $\mathbf{w}_j = \lambda \mathbf{a}_j$, $b_j \leftarrow \lambda b_j$. ∎

Remark 8S.29: the output constant

The handwritten statement leaves the constant implicit. The proof approximates $f - f(\mathbf{0})$ and restores $f(\mathbf{0})$ as the output bias a_0. Encoding the constant as an extra hidden unit changes the unit count to $N + 1$; the two conventions should not be mixed silently.

What the rate means. The squared error is $O(N^{-1})$, so the L^2 error is $O(N^{-1/2})$. With C_1 fixed, the exponent does not depend on d; this is the sense in which the result avoids a dimension-dependent exponent. However, the Barron constant of a target can itself grow with d, so the theorem does not cover arbitrary high-dimensional functions with a uniform constant. It also says nothing about the sample size n or about the success of gradient descent.

	Universal approximation	Barron approximation
Target class	continuous functions on K	controlled Fourier cost Γ_{C_1}
Error criterion	uniform norm	normalized squared L^2 on B_1^d
Conclusion	arbitrarily accurate	explicit c/N rate
Width	no quantitative bound	quantitative dependence on N

Cybenko answers whether representation is possible; Barron answers how efficiently it is possible for a structured class. Neither replaces the statistical and optimization terms of Theorem 8S.6. The handout's lecture material ends at the statement of Theorem 8S.28; the convex-hull lemma, the proof and the Rademacher bounds are in Chapter 8.

8 Exercises

Exercise 8S.1 (recovering the decomposition)

Starting from $R(\widehat{f}_S) - \inf_{\mathcal{F}_\delta} R$, derive the three terms of (5). Which two are bounded in absolute value by $\Delta_S(\mathcal{F}_\delta)$?
Solution. Add and subtract $\widehat{R}_S(\widehat{f}_S)$ and $\inf_{\mathcal{F}_\delta} \widehat{R}_S$:

$$R(\widehat{f}_S) - \inf R = [R(\widehat{f}_S) - \widehat{R}_S(\widehat{f}_S)] + [\widehat{R}_S(\widehat{f}_S) - \inf \widehat{R}_S] + [\inf \widehat{R}_S - \inf R].$$

The first (the gap of $\widehat{f}_S$) and the third (Steps 1–2 of the proof of Theorem 8S.6) are bounded by Δ_S; the middle one is ε_{opt}.

Exercise 8S.2 (fixed versus learned)

Why can the variance calculation of Theorem 8S.7 not be reused for $\widehat{f}_S$?
Solution. It needs $Z_1, \dots, Z_n$ iid with mean $R(f)$, which holds only when f is chosen independently of S. For $\widehat{f}_S$ every $Z_i = \ell(\widehat{f}_S(X_i), Y_i)$ depends on the whole sample, and the memorizing predictor of Example 8S.8 shows the gap can stay at $\frac{1}{2}$ for all n.

Exercise 8S.3 (hinge ordering)

In Example 8S.15, why is the ordering $R_{\text{hinge}}(g_B) < R_{\text{hinge}}(g_A)$ consistent with $R_{0\text{–}1} \le R_{\text{hinge}}$?
Solution. The pointwise bound only compares each score's two risks: $0 \le 0.9$ for g_A and $0.01 \le 0.011$ for g_B. It says nothing about how the risks of *different* scores are ordered.

Exercise 8S.4 (fixed width)

Why is $\mathcal{G}_\sigma$ (all widths) a linear space while the class of width exactly N need not be?
Solution. The sum of two width-N networks is a width-$2N$ network, which may not be representable with N units: for $N = 1$ and the logistic σ, $\sigma(t) + \sigma(t-5)$ is not of the form $\alpha\sigma(wt+\theta)$, since the latter is constant or has exactly one inflection point, while the second derivative of the sum changes sign three times (it is positive at $t = 0$ and $t = 4$, negative at $t = 1$ and $t = 6$). Allowing all N closes the class under addition.

Exercise 8S.5 (where each ingredient enters)

Point to where $L \neq 0$, $\mu \neq 0$ and the discriminatory property enter the density proof.
Solution. $L \neq 0$ with $L|_{\mathcal{G}_\sigma} = 0$ is Step 2 (Hahn–Banach, Theorem 8S.21(i)); $\mu \neq 0$ follows from $L \neq 0$ through Riesz (ii); discrimination is used in Step 3 to conclude $\mu = 0$.

Exercise 8S.6 (four symbols)

Distinguish n, N, N^{-1} and $N^{-1/2}$.
Solution. n is the sample size (statistical error, e.g. $O_P(n^{-1/2})$ for a fixed predictor); N is the hidden width; N^{-1} is Barron's rate for the *squared* L^2 error; $N^{-1/2}$ is the corresponding rate for the L^2 error itself.